

Brian M. Howell

computational optimization | numerical methods | high performance computing
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Summary

I am an engineer interested in high-performance numerical simulation/optimization/linear algebra and its applications in physics and AI/ML.

Professional Experience

Apple, Cupertino, CA.

Feb 2024 - present

Optimization/ML Research Engineer

- **Platform Architecture:** I work on a team focusing on parallel/distributed high-performance computing tools for internal use.
- **AI:** I lead the infrastructure development of an internal AI engine for accelerating forward-modeling workflows for internal use.
- **Optimization:** I also led the development of an internal optimization engine for large-scale distributed computing tasks related to high performance computing and machine learning.
- **GPU kernel engineering:** Parallel computing via GPUs and MPI, low-level kernel development and performance optimization, large-scale distributed computing for numerical optimization, machine learning for low-rank accelerated modeling.

Google X, the moonshot factory, Mountain View, CA.

Jan 2022 - Dec 2022

AI Resident:

- **My project** aimed to bring modern computing tools for materials optimization to a very large industry. My colleague and I cracked the problem with geometric/thermodynamic + convex modeling/optimization.
- **Machine Learning/Optimization:** Gaussian processes + Bayesian optimization, deep learning, convex optimization
- **Physics Simulation/Modeling:** Discrete element method, convex geometry
- **Hardware:** Sensor development and data processing, high-throughput experimentation, feedback control systems for complex fluid flow
- **Publicly Available Output:** Three patents granted (one as lead inventor)

Lawrence Livermore National Lab., Livermore, CA.

June 2017 - Jan 2022

Staff Scientist:

- **My work** at LLNL was primarily focused on materials development & optimization for 3D printing
- **Software/Simulation:** Controllers, sensors, toolpath generation and optimization, digital twins for additive manufacturing
- **Hardware/Chemical:** Hardware integration, CAD modeling & design, chemical formulation
- **Testing:** Rheology & UV kinetics, mechanical (Instron), Scanning Electron Microscope (SEM)
- **Publicly Available Output:** Two patents, one publication, work featured in [Advanced Science News](#)

Skills

Programming Tools: C/C++, Python, CUDA, METAL, OpenMP, MPI, PyTorch, JAX, MLX, Git, Linux

Computational Methods: Numerical Methods/Optimization/Linear Algebra, Machine Learning, Parallel Computing, Distributed Computing

Education

UC Berkeley

PhD/MS in Computational/Mechanical Engineering

2019-2024

Dissertation: *Simulation-Informed Optimization and Machine Learning for Advanced Manufacturing*

Advisor: Prof. Tarek Zohdi

Brigham Young University

Emphasis: Control Theory

BS in Chemical Engineering

2013-2017